Learnings & Reflections Document

1. Team Reflection

We worked by assigning roles based on periodic self evaluations of our work. It wasn't a rigid divide of work. We were shifting roles daily based on the most immediate technical bottleneck. Because the project faced constant hardware setbacks, we were often spending hours together just debugging a single short section code.

To me, our ability to stay composed during system failures was our biggest strength. When the file system corrupted or we hit the KCKC noise, we jumped into fixing it, and the fact that we were doing this throughout every single code that we tested, is what helped our project work (even though it wasn't all the way we had hoped).

Moreover, there was cooperation between us so the whole process was smooth. When the ferrofluid wasn't arriving in week three, we pivoted to researching and implementing the Chladni patterns concept quickly and without panicking and giving up. Looking back at what we could have improved, I think we underestimated the last minute issues, and how the troubleshooting process would be such a blind search sometimes. I also understand that taking feedback from faculty throughout would have helped us avoid a lot of the issues that we faced last minute.

2. Technical Reflection

The main struggle was managing the I2C bus stability once we added the NeoPixel ring and the PAM8403 amplifier. The system was prone to hanging because of electrical interference, which showed up as random garbled symbols in our serial monitor and forced us to learn how to clear the FIFO registers manually. Every time the heart rate peaked, the surge in power draw would occasionally crash the I2C communication, requiring a hard reset of the logic. Overall, The biggest technical challenge of the project was not a single component, it was getting multiple components to coexist on the same power supply. Individually, the pulse sensor worked. The NeoPixel worked. The speaker worked. Together, running all of them simultaneously, the system kept failing. The electromagnet and motor drew too much current when combined with everything else, and the power supply couldn't keep up. We eventually excluded both from the final demo and used a Buck Converter to stabilise the remaining components. That decision to cut the two components from the final demo was our only option even though it was a hard step to take.

We learned the importance of power isolation and why a common ground is mandatory when mixing logic and high-current loads. Software-wise, we learned how to manage FIFO buffers in MicroPython to keep the sensor data from overflowing and causing a panic. We also learned how to map real-time biological data into a pentatonic frequency, ensuring that the output wasn't just noise, but a coherent musical response to the body. We also learned that floating connections are not good enough for testing. The PAM8403 amplifier was barely functional during week two testing because the wires connected to it weren't fixed. The results were unreliable, which made it impossible to know whether the component worked. You have to commit to a connection, tape it, clip it, solder it before you can trust what the test is telling you. The Chladni experiment taught us something we didn't expect to learn: that acoustic physics produces visible geometry with almost no engineering required. A metal plate, a speaker, a frequency, and fine powder is enough. That simplicity was surprising and genuinely interesting. We wasted time assuming our code was the problem when the issue was actually "noise" from the LEDs. We eventually fixed this by using a Buck Converter to provide a clean 5V rail, which finally stopped the ESP32 from crashing every time the pixels flashed. We also had to deal with a corrupted main.py file due to a laptop crash, which taught us the importance of frequent backups and serial communication stability. The most important hardware lesson was about power. A component that works fine on its own can cause a voltage drop that collapses everything else when it's added to a shared supply. Designing the power architecture, calculating worst-case current draw for every component and choosing a supply that exceeds it with margin has to happen before anything else. We learned this the hard way when our circuit experienced a short. Fortunately we were able to recover from it and no major damage was caused.

3. Design Reflection

Conceptually, the experience matched our intention. We wanted to turn a heartbeat into a direct sensory experience that felt alive. However, the physical build ended up feeling more like a prototype than a finished product because the sensor calibration was quite sensitive to how the user touched it.While the light sensing was stable, the interaction felt "technical" rather than "organic" because of the physical precision required. To get a consistent pulse reading, the user had to hold their finger at a very specific angle and pressure. If they moved even slightly, the I2C bus would struggle to track the peak-detection. This meant the user was often more focused on properly touching the sensor than on the experience of the light and sound.
The biggest design flaw was the internal architecture of the record player box. While the Blender-modeled enclosure looked professional, it was definitely something that we could have taken to building with mdf. I would also redesign the housing with internal "bulkheads" to physically separate the PAM8403 amplifier and the 12V-to-5V Buck Converter from the I2C data lines. This would have likely prevented the "KCKC" noise that occurred every time the speaker drew a spike of current, which we believe was causing the system to hang during the demo. Additionally, I would look at a more compact, organized structure for the record player box, as a lot of the hollow space inside it went unused due to the changes we made in the setup later. I realize that despite the structure of the box was not the core focus of the project, I could have given more time to coming up with a structure for it that felt more carefully thought through.
Most users were surprised by the Pentatonic Mapping, they were intrigued when we told them that their pulse rate was being converted mathematically, into a series of beats in the pentatonic scale. However, we noticed that without a better physical "lock" for the finger, the interaction felt fragile; users were afraid to move for fear of breaking the connection. Looking back, I would have made a more ergonomic touch system for the pulse rate sensor. Users also mentioned

that the 12V power adapter made the project feel less "portable," so a future version would definitely benefit from an integrated battery management system to make the installation more self-contained.

4. Personal Reflection

I learned that in a high-pressure environment, technical failure feels much more personal than it actually is. This module forced me to realize that being a designer means showing up and trying to fix the problem even when you feel like you're being watched through a lens of skepticism. I've gained more technical knowledge from the things that broke than from the things that worked. Through most of the process, I struggled with a constant sense of self-doubt. Throughout this module, it often felt like there was no room for mistakes, and that every technical hurdle was a sign that I didn't belong in the class. The component issues and the code crashes weren't just bugs to me; they felt like proof that I wasn't grasping the material, which made the process incredibly isolating. That being said, I am proud that I didn't let the technical setbacks make me quit. We built a custom I2C recovery routine and a 3D-printed enclosure from scratch under intense pressure. I am proud that I advocated for our project and kept pushing to find solutions, like the Buck Converter, when it would have been easier to just give up. I am also proud of my partner Twisha, in an environment that felt increasingly hostile we didn't just share the workload; we shared the emotional weight of trying to prove ourselves to the faculty, and back to ourselves too. To be honest, I am carrying a lot of disappointment that the system didn't run perfectly for the final submission. I cared deeply about this module, despite it maybe not seeming that way to Nayan, and I really wanted this project to be the moment I finally proved my capability and ideas.It's hard to accept that after weeks of the developing the project, the final demo was flawed. But I know the work we did was real, even if the final result didn't show it in the way I hoped.